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CSE 13S Spring 2021
Assignment 1: Left, Right and Center
Design Document

New changes are in red

Purpose

In the game *Left, Right, and Center*, there are 2 to 14 players that each have $3 and three dice with 6 faces each. 3 of the faces are a •, 1 is an L (left), 1 is an R (right), and 1 is a C (center). Player 1 rolls 3 dice if he or she has $3 or more. A player rolls 2 dice if he or has $2 and 1 die if he or she has $1. Players pass if they have no money.
For each die:

- If the player rolls L, he gives $1 to the player on his left.
- If the player rolls R, he gives $1 to the player on his right.
- If the player rolls C, he puts the $1 in the pot in the center.
- If the player rolls •, he ignores it.

Then move to the next player to the right. Repeat until only one player has any money. That remaining player wins the pot in the center.

I will implement this game by using the random() function to simulate rolling a dice. The range will be from 0-5 to represent the six faces of the dice.

The inputs to our program are:

1. The random seed
2. The number of players (must be from 1 to 14)

Expected outputs:

1. Error message if the seed is invalid (must be a number greater than 1)
2. Error message if the number of players is invalid (must be a number from 1 to 14)
3. A correct simulation of the game if the random seed and number of player inputs are valid

The purpose of the program is to implement the *Left, Right, and Center* game according to the established rules. When using the same random seed and the same number of players, the game simulation will always be the same.

Layout/Structure

- Before the game starts, the user is prompted to enter a valid seed that is greater than or equal to 1 and a number of players from 2 to 14.

- The program will print an error message and end if the user enters an invalid seed or invalid number of players.
- The game runs in a while loop that breaks when there is only 1 player left in the game.
- There is a for loop inside the while loop to loop through the players of the game.
- There is a nested for loop inside that for loop to determine how many players are still in the game.
- Each player rolls a number of times determined by the minimum between 3 and how many dollars he or she has.
- There is a for loop inside the player for loop that simulates the rolling of the dice.
- At the end of the program, there is a for loop to determine which player has money left and is the winner of the game.

Functionality

- The left function returns the position/index of the player to the left of the current player. This allows us to increment the bank of the player to the left.
- The right function returns the position/index of the player to the right of the current player. This allows us to increment the bank of the player to the right.
- The main function contains the code for the game. It includes:
    - the enumeration of the die faces - assigns the values 0,1,2 to LEFT, RIGHT, and CENTER
    - the prompts to read the seed and the number of players - reads an integer greater than or equal to 1 for the seed and a number from 2 to 14 for the number of players
    - the error handling for invalid inputs - tells the user that the input was invalid and exits the program
    - the while loop for the game - this is where the game runs until there is only 1 player left
    - The for loop for looping through the players - each player has a turn to roll if he or she has money
    - the for loop for counting how many players are still in the game - counts how many players still have money left
    - The for loop for each player's rolls - each player rolls for the number of dollars they have or 3 times if they have $3 or more dollars
    - The for loop for determining which player wins the game - the one player that has money left at the end of the game is the winner

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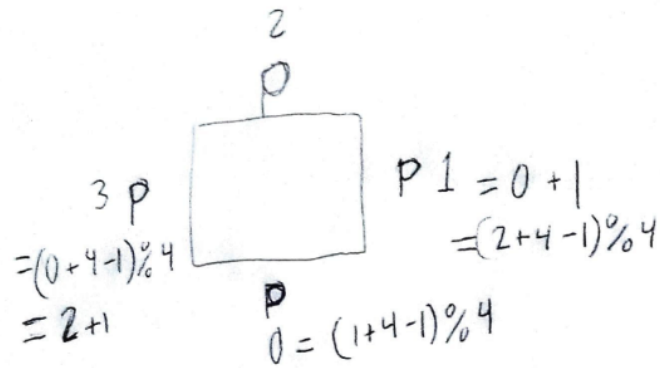
TOP LEVEL:

The top level design of the code is given by the following pseudocode:

```
Left function:
    Return ((position + players - 1) % players)

Right function:
    Return ((position + 1) % players)

Main function:
    Enumerate pass, left, right, center with type faces
    die array of type faces = { LEFT, RIGHT, CENTER, PASS, PASS }
    dollars array = { 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3 }
    Declare integer seed
    Declare integer number of players
    Declare integer roll state
    Pot = 0

    Print “Random seed: “
    Read the seed
    If number of items read < 1 or seed < 1:
        Print “Pseudorandom seed must be non-negative (<seed>)”
        Print new line
        End program

    Print “How many players? “
    Read the number of players
    If number of items read < 1 or number of players  is not > 1 and <= 14:
        Print(“Number of players must be from 1 to 14.”)
        Print new line
        End program
    Set random seed

    Alive = number of players
    While alive > 1:

        For each player:
            Roll state = 0
            Alive = 0
            For each player:
                If player’s dollars > 0
                    Increment alive
            If alive == 1:
                Break
            If player’s dollars > 0:
                Print “<philosopher> rolls…”
```

```
                Roll state = 1
            Declare integer roll count
            Space count = 0
            If player's dollars > 3:
                Roll count = 3
            Else:
                Roll count = player's dollars

            For roll count times:
                Roll num = random number % 6
                If player rolled LEFT:
                    Decrement player's dollars
                    Increment left player's dollars
                    Print "gives $1 to <left philosopher>"

                If player rolled RIGHT:
                    Decrement player's dollars
                    Increment right player's dollars
                    Print "gives $1 to <right philosopher>"

                If player rolled CENTER:
                    Decrement player's dollars
                    Increment pot
                    Print "puts $1 in the pot"

                If player rolled PASS
                    Print "gets a pass"

                If space count < roll count - 1:
                    Print " "

                Increment space count

            If roll state == 1:
                Print new line

For each player:
    If player's dollars > 0:
        Print "<philosopher> wins the $<pot> pot with $<player's dollars> left in
        the bank"
        Print new line
End program
```

PAPER DRAFT AND WORK



P0 roll ×3

P1 roll ×3

P2 roll ×3

P3 roll ×3

num rolls = num $

at start: num alive = num players

if player x has $0, num alive -=1

if player with $0 is given $1, num alive +=1

if num alive = 1, end game